

1. Let $A = [a_{ij}]_{2 \times 2}$, where $a_{ij} \neq 0$ for all i, j and $A^2 = I$. Let a be the sum of all diagonal elements of A and $b = |A|$. Then $3a^2 + 4b^2$ is equal to

[2023 (06 Apr Shift 1)]

- (1) 4
(2) 14
(3) 7
(4) 3

2. Let P be a square matrix such that $P^2 = I - P$. For $\alpha, \beta, \gamma, \delta \in \mathbb{N}$, if $P^\alpha + P^\beta = \gamma I - 29P$ and $P^\alpha - P^\beta = \delta I - 13P$, then $\alpha + \beta + \gamma - \delta$ is equal to

[2023 (06 Apr Shift 2)]

- (1) 18
(2) 40
(3) 22
(4) 24

3. Let $\begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$. If $|\text{adj}(\text{adj}(\text{adj}(2A)))| = (16)^n$, then n is equal to

[2023 (08 Apr Shift 1)]

- (1) 8
(2) 10
(3) 9
(4) 12

4. Let $P = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$, $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $Q = PAP^T$. If $P^T Q^{2007} P = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ then $2a + b - 3c - 4d$ is equal to

[2023 (08 Apr Shift 1)]

- (1) 2004
(2) 2005
(3) 2007
(4) 2006

5. If $A = \begin{bmatrix} 1 & 5 \\ \lambda & 10 \end{bmatrix}$, $A^{-1} = \alpha A + \beta I$ and $\alpha + \beta = -2$, then $4\alpha^2 + \beta^2 + \lambda^2$ is equal to :

[2023 (08 Apr Shift 2)]

- (1) 12
(2) 19
(3) 14
(4) 10

6. If A is a 3×3 matrix and $|A| = 2$, then $|3 \text{adj}(|3A|A^2)|$ is equal to

[2023 (10 Apr Shift 1)]

- (1) $3^{12} \cdot 6^{11}$
(2) $3^{12} \cdot 6^{10}$
(3) $3^{10} \cdot 6^{11}$
(4) $3^{11} \cdot 6^{10}$

7. If $A = \frac{1}{5!6!7!} \begin{bmatrix} 5! & 6! & 7! \\ 6! & 7! & 8! \\ 7! & 8! & 9! \end{bmatrix}$, then $|\text{adj}(\text{adj}(2A))|$ is equal to

[2023 (10 Apr Shift 2)]

- (1) 2^{20}
(2) 2^8
(3) 2^{12}
(4) 2^{16}

8. Let $A = \begin{bmatrix} 0 & 1 & 2 \\ a & 0 & 3 \\ 1 & c & 0 \end{bmatrix}$, where $a, c \in \mathbb{R}$. If $A^3 = A$ and the positive value of a belongs to the interval $(n-1, n]$, where $n \in \mathbb{N}$, then n is equal to ____.

[2023 (11 Apr Shift 1)]

9. Let $A = \begin{bmatrix} 1 & \frac{1}{51} \\ 0 & 1 \end{bmatrix}$. If $B = \begin{bmatrix} 1 & 2 \\ -1 & -1 \end{bmatrix} A \begin{bmatrix} -1 & -2 \\ 1 & 1 \end{bmatrix}$, then the sum of all the elements of the matrix $\sum_{n=1}^{50} B^n$ is equal to

[2023 (12 Apr Shift 1)]

- (1) 75
(2) 125
(3) 50
(4) 100

10. The number of symmetric matrices of order 3, with all the entries from the set $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$ is

[2023 (13 Apr Shift 1)]

- (1) 6^{10}
(2) 10^6
(3) 9^{10}
(4) 10^9

11. Let $B = \begin{bmatrix} 1 & 3 & \alpha \\ 1 & 2 & 3 \\ \alpha & \alpha & 4 \end{bmatrix}$, $\alpha > 2$ be the adjoint of a matrix A and $|A| = 2$. Then $\begin{bmatrix} \alpha & -2\alpha & \alpha \end{bmatrix} B \begin{bmatrix} \alpha \\ -2\alpha \\ \alpha \end{bmatrix}$ is equal to

[2023 (13 Apr Shift 1)]

- (1) 0
(2) 16
(3) -16
(4) 32

12. Let for $A = \begin{bmatrix} 1 & 2 & 3 \\ \alpha & 3 & 1 \\ 1 & 1 & 2 \end{bmatrix}$, $|A| = 2$. If $|2 \operatorname{adj}(2 \operatorname{adj}(2A))| = 32^n$, then $3n + \alpha$ is equal to

[2023 (13 Apr Shift 2)]

- (1) 9
(2) 11
(3) 12
(4) 10

13. Let the determinant of a square matrix A of order m be $m - n$, where m and n satisfy $4m + n = 22$ and $17m + 4n = 93$. If $\det(n \operatorname{adj}(\operatorname{adj}(mA))) = 3^a 5^b 6^c$, then $a + b + c$ is equal to

[2023 (15 Apr Shift 1)]

- (1) 84
(2) 96
(3) 101
(4) 109

ANSWER KEYS

1. (1) 2. (4) 3. (2) 4. (2) 5. (3) 6. (4) 7. (4) 8. (2)
9. (4) 10. (2) 11. (3) 12. (2) 13. (2)

1. (1)

Let,

$$A = \begin{bmatrix} p & q \\ r & s \end{bmatrix}$$

$$A^2 = \begin{bmatrix} p & q \\ r & s \end{bmatrix} \begin{bmatrix} p & q \\ r & s \end{bmatrix} = \begin{bmatrix} p^2 + qr & pq + qs \\ rp + rs & qr + s^2 \end{bmatrix}$$

Now given, $A^2 = I$

$$\Rightarrow \begin{bmatrix} p^2 + qr & pq + qs \\ rp + rs & qr + s^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Now on comparing both side we get,

$$\Rightarrow p^2 + qr = 1, q(p + s) = 0$$

$$\text{And } r(p + s) = 0, qr + s^2 = 1$$

Now on solving above relation we get,

$$q \neq 0 \Rightarrow p + s = 0 \Rightarrow a = 0$$

$$\text{And } b = |A| = ps - qr = -p^2 - qr = -1 (\because s = -p)$$

$$\therefore 3a^2 + 4b^2 = 4$$

2. (4)

Given,

$$P \text{ be a square matrix such that } P^2 = I - P,$$

Now solving $P^2 = I - P$ we get,

$$\Rightarrow P^4 = (I - P)(I - P)$$

$$\Rightarrow P^4 = I + P^2 - 2P$$

$$\Rightarrow P^4 = I + I - P - 2P = 2I - 3P$$

$$\text{Now solving, } P^4 \cdot P^2 = P^6 = 2I - 5P + 3P^2$$

$$\Rightarrow P^6 = 2I - 5P + 3(I - P) = 5I - 8P \dots (i)$$

$$\text{Similarly } P^8 = 13I - 21P \dots (ii)$$

Now adding (ii)+(i) we get,

$$P^8 + P^6 = 18I - 29P$$

Now subtracting (ii)-(i) we get,

$$P^8 - P^6 = 8I - 13P$$

Now comparing with $P^\alpha - P^\beta = \delta I - 13P$ & $P^\alpha + P^\beta = \gamma I - 29P$ we get,

$$\alpha = 8, \beta = 6, \gamma = 18, \delta = 8$$

$$\text{Hence, } \alpha + \beta + \gamma - \delta = 8 + 6 + 18 - 8 = 24$$

3. (2)

Given that $A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$ and $|\text{adj}(\text{adj}(\text{adj}2A))| = 16^n$.

We know that $|\text{adj}A| = |A|^{n-1}$ and $|\text{adj}(\text{adj}A)| = |A|^{(n-1)^2}$

Similarly, $|\text{adj}(\text{adj}(\text{adj}A))| = |A|^{(n-1)^3}$ where n is the order of the square matrix.

Let us find $|A|$.

$$|A| = 2(2 \times 2 - (-1)(-1)) - 1(1 \times 2 - 0(-1)) + 0(1 \times (-1) - 0 \times 2)$$

$$= 2(3) - (2) + 0 = 4$$

$$\Rightarrow |2A|^{(n-1)^3} = |2A|^{(3-1)^3}$$

$$= |2A|^8 = (2^3)^3 |A|^8$$

$$\Rightarrow 2^{24} \times 4^8 = 16^n$$

$$\Rightarrow 16^6 \times 16^4 = 16^n$$

$$\Rightarrow 16^{10} = 16^n$$

$$\Rightarrow n = 10$$

Therefore, the value of n is 10.

4. (2)

Given:

$$P = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$$

$$\therefore PP^T = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow PP^T = I$$

Now,

$$P^T(PAP^T)^{2007}P = P^T \underbrace{(PAP^T)(PAP^T)(PAP^T) \dots (PAP^T)}_{2007 \text{ times}} P$$

$$\Rightarrow P^T(PAP^T)^{2007}P = A^{2007}$$

Now,

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$$

$$\vdots$$

$$\vdots$$

$$\vdots$$

$$\vdots$$

$$\vdots$$

$$A^{2007} = \begin{bmatrix} 1 & 2007 \\ 0 & 1 \end{bmatrix}$$

So,

$$\Rightarrow P^T(PAP^T)^{2007}P = \begin{bmatrix} 1 & 2007 \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow P^TQP = \begin{bmatrix} 1 & 2007 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

So,

$$a = 1, b = 2007, c = 0, d = 1$$

$$2a + b + 3c - 4d = 2 + 2007 - 4 = 2005$$

5. (3)

Given,

$$A = \begin{bmatrix} 1 & 5 \\ \lambda & 10 \end{bmatrix}, A^{-1} = \alpha A + \beta I \text{ and } \alpha + \beta = -2,$$

Now solving by using characteristic equation, we get

$$\begin{vmatrix} A - kI \end{vmatrix} = 0 \Rightarrow \begin{vmatrix} 1-k & 5 \\ \lambda & 10-k \end{vmatrix} = 0$$

$$\Rightarrow k^2 - 11k + 10 - 5\lambda = 0$$

$$\Rightarrow A^2 - 11A + (10 - 5\lambda)I = 0 \text{ \{by putting } k = A\}}$$

$$\Rightarrow A^{-1} = \frac{1}{10-5\lambda}(-A + 11I)$$

Now on comparing with $A^{-1} = \alpha A + \beta I$ we get,

$$\Rightarrow \alpha = -\frac{1}{10-5\lambda}, \beta = \frac{11}{10-5\lambda}$$

And given,

$$\alpha + \beta = -2 \Rightarrow \lambda = 3, \alpha = \frac{1}{5}, \beta = \frac{-11}{5}$$

$$\text{Hence, the value of } 4\alpha^2 + \beta^2 + \lambda^2 = 14$$

6. (4)

We need to find the value of $|3adj(3A|A^2)|$ We know that $|kA| = k^n|A|$ where n is the order of the matrix and k is a constant.

$$\Rightarrow |3A| = 3^3(2)$$

$$\Rightarrow |3adj(3A|A^2)| = 3^3|adj((3^3 \cdot 2)A^2)|$$

We know that $adj(kA) = k^{n-1}adj(A)$

$$= 3^3|(3^3 \cdot 2)^2 adj(A^2)|$$

$$\text{Now we know that } |adjA| = |A|^{n-1}$$

$$= 3^3((3^3 \cdot 2)^2)^3 |adj(A^2)|$$

$$= 3^3(3^3 \cdot 2)^6 |A^2|^{3-1}$$

$$= 3^3(3^3 \cdot 2)^6 (2)^4$$

$$= 2^{10} \cdot 3^{21}$$

$$\text{Hence, the required answer is } 2^{10} \cdot 3^{21} = 6^{10} \cdot 3^{11}$$

7. (4)

Given,

$$A = \frac{1}{5!6!7!} \begin{bmatrix} 5! & 6! & 7! \\ 6! & 7! & 8! \\ 7! & 8! & 9! \end{bmatrix}$$

$$\Rightarrow |A| = \frac{1}{5!6!7!} \begin{vmatrix} 5! & 6! & 7! \\ 6! & 7! & 8! \\ 7! & 8! & 9! \end{vmatrix}$$

$$\Rightarrow |A| = \frac{1}{5!6!7!} 5! \times 6! \times 7! \begin{vmatrix} 1 & 6 & 42 \\ 1 & 7 & 56 \\ 1 & 8 & 72 \end{vmatrix}$$

$$\Rightarrow |A| = 2$$

Now using the property of adjoint of matrix we get,

$$|adj(adj(A))| = |A|^{(n-1)^2} \text{ \& } |kA| = k^n|A|, \text{ where } n \text{ is order of square matrix,}$$

Now using the above two formula we get,

$$|adj(adj(2A))| = |2A|^4 = 2^{12}|A|^4$$

$$\Rightarrow |adj(adj(2A))| = 2^{12} \cdot 2^4$$

$$\Rightarrow |adj(adj(2A))| = 2^{16}$$

8. (2)

Given,

$$A = \begin{bmatrix} 0 & 1 & 2 \\ a & 0 & 3 \\ 1 & c & 0 \end{bmatrix}$$

$$\Rightarrow A^2 = \begin{bmatrix} 0 & 1 & 2 \\ a & 0 & 3 \\ 1 & c & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 2 \\ a & 0 & 3 \\ 1 & c & 0 \end{bmatrix}$$

$$\Rightarrow A^2 = \begin{bmatrix} a+2 & 2c & 3 \\ 3 & a+3c & 2a \\ ca & 1 & 2+3c \end{bmatrix}$$

Now finding,

$$A^3 = \begin{bmatrix} a+2 & 2c & 3 \\ 3 & a+3c & 2a \\ ca & 1 & 2+3c \end{bmatrix} \begin{bmatrix} 0 & 1 & 2 \\ a & 0 & 3 \\ 1 & c & 0 \end{bmatrix}$$

$$\Rightarrow A^3 = \begin{bmatrix} 2ca+3 & \alpha+2+3c & 2a+4+6c \\ a^2+3ca+2a & 3+2ac & 6+3a+9c \\ a+2+3c & ca+2c+3c^2 & 2ca+3 \end{bmatrix}$$

Now equating $A^3 = A$ and comparing both side we get,

$$2ca+3=0 \Rightarrow c = \frac{-3}{2a}$$

$$\text{And } a+2+3c=1$$

$$\Rightarrow a+2+3\left(\frac{-3}{2a}\right)=1$$

$$\Rightarrow a+1-\frac{9}{2a}=0$$

$$\Rightarrow 2a^2+2a-9=0$$

$$\Rightarrow a = \frac{-2+\sqrt{4+4\times 9\times 2}}{4} = \frac{-2+\sqrt{76}}{4} \approx \frac{6.7}{4} \approx 1.4$$

Hence, $a \in (1, 2]$ So, on comparing with $a \in (n-1, n]$ we get $n=2$

9. (4)

Given,

$$A = \begin{bmatrix} 1 & \frac{1}{51} \\ 0 & 1 \end{bmatrix},$$

$$B = \begin{bmatrix} 1 & 2 \\ -1 & -1 \end{bmatrix} A \begin{bmatrix} -1 & -2 \\ 1 & 1 \end{bmatrix}$$

So, $B = MAN$

$$\text{Now let } M = \begin{bmatrix} 1 & 2 \\ -1 & -1 \end{bmatrix} \& N = \begin{bmatrix} -1 & -2 \\ 1 & 1 \end{bmatrix}$$

Now solving,

$$\begin{matrix} (M) & & (N) \\ \begin{bmatrix} 1 & 2 \\ -1 & -1 \end{bmatrix} & \begin{bmatrix} -1 & -2 \\ 1 & 1 \end{bmatrix} & = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \end{matrix}$$

$$\therefore MN = I = NM$$

Now using, $B = MAN$ we get

$$B^n = (MAN)^n = (MAN)(MAN) \dots (MAN)$$

$$\Rightarrow B^n = \underbrace{MANMAN \dots MAN}_I \underbrace{NM}_I \underbrace{AN}_I$$

$$\Rightarrow B^n = \underbrace{MA \dots MA}_I \dots AN = MA^n N$$

$$\text{Now, using } A = \begin{bmatrix} 1 & \frac{1}{51} \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & \frac{1}{51} \\ 0 & 0 \end{bmatrix} = I + E$$

$$\text{Now finding, } E^2 = \begin{bmatrix} 0 & \frac{1}{51} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & \frac{1}{51} \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

 $\Rightarrow E^2 = 0$, so all higher power will also be zero,

$$\therefore A^n = (I + E)^n = I + nE + \underbrace{{}^nC_2 E^2}_0 + \underbrace{{}^nC_3 E^3}_0 + \underbrace{{}^nC_4 E^4}_0 + \dots$$

$$\Rightarrow A^n = I + nE$$

$$\Rightarrow A^n = \begin{bmatrix} 1 & \frac{n}{51} \\ 0 & 1 \end{bmatrix}$$

$$\text{So, } B^n = MA^n N = \begin{bmatrix} 1 & 2 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} 1 & \frac{n}{51} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} -1 & -2 \\ 1 & 1 \end{bmatrix} \Rightarrow B^n = \begin{bmatrix} 1 & \frac{n}{51} + 2 \\ -1 & \frac{-n}{51} - 1 \end{bmatrix} \begin{bmatrix} -1 & -2 \\ 1 & 1 \end{bmatrix}$$

$$\Rightarrow B^n = \begin{bmatrix} 1 + \frac{n}{51} & \frac{n}{51} \\ \frac{-n}{51} & 1 - \frac{n}{51} \end{bmatrix}$$

$$\sum_{n=1}^{50} B^n = \begin{bmatrix} 50 + \frac{50 \cdot 51}{2 \cdot 51} & \frac{50 \cdot 51}{2 \cdot 51} \\ \frac{-50 \cdot 51}{2 \cdot 51} & 50 - \frac{50 \cdot 51}{2 \cdot 51} \end{bmatrix} = \begin{bmatrix} 75 & 25 \\ -25 & 25 \end{bmatrix}$$

$$\therefore \text{Sum} = 100$$

10. (2)

Given that $A = \{0, 1, 2, 3, \dots, 8, 9\}$

We need to find the number of symmetric matrices.

We know that a matrix is symmetric if $A^T = A$.

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Now a_{11}, a_{22}, a_{33} can have any digit from 0 to 9.Number of ways this can be done is $10 \times 10 \times 10 = 10^3$ Now $a_{12} = a_{21}$. So number of ways this can be done is 10×1 .Similarly $a_{13} = a_{31}$, So number of ways this can be done is 10×1 .Also, $a_{23} = a_{32}$. So number of ways this can be done is 10×1 .

$$\therefore \text{Total symmetric matrix} = 10^3 \times 10 \times 10 \times 10 = 10^6$$

Hence this is the correct option.

11. (3)

We have been given that $B = \begin{bmatrix} 1 & 3 & \alpha \\ 1 & 2 & 3 \\ \alpha & \alpha & 4 \end{bmatrix}, \alpha > 2$

And $\text{adj}(A) = B, |A| = 2$

$\Rightarrow |\text{adj}(A)| = |B|$

We know that $|\text{adj}A| = |A|^{n-1}$

$$\Rightarrow 2^2 = (8 - 3\alpha) - 3(4 - 3\alpha) + \alpha(-\alpha)$$

$$\Rightarrow \alpha^2 - 6\alpha + 8 = 0$$

$$\Rightarrow (\alpha - 4)(\alpha - 2) = 0$$

$$\alpha = 4, 2 \text{ but } \alpha > 2 \text{ so } \boxed{\alpha = 4}$$

$$\text{Now } \begin{bmatrix} \alpha & -2\alpha & \alpha \end{bmatrix} B = \begin{bmatrix} 1 & 3 & 4 \\ 1 & 2 & 3 \\ 4 & 4 & 4 \end{bmatrix} \begin{bmatrix} 4 \\ -8 \\ 4 \end{bmatrix}$$

$$= \begin{bmatrix} 12 & 12 & 8 \end{bmatrix} \begin{bmatrix} 4 \\ -8 \\ 4 \end{bmatrix} = 48 - 96 + 32 = -16$$

Hence this is the correct option.

12. (2)

$$\text{Given that } A = \begin{bmatrix} 1 & 2 & 3 \\ \alpha & 3 & 1 \\ 1 & 1 & 2 \end{bmatrix}$$

$$\Rightarrow |A| = 2$$

$$\Rightarrow 1(6 - 1) - \alpha - 7 = 2$$

$$\Rightarrow -\alpha - 2 = 2$$

$$\Rightarrow \alpha = -4$$

$$\Rightarrow |2 \text{adj}(2 \text{adj}(2A))| = 32^n$$

We know that $|kA| = k^n |A|$, $\text{adj}(A) = |A|^{(n-1)^2}$ and $\text{adj}(kA) = k^{n-1} \text{adj}(A)$.

$$= 2^3 |\text{adj}(2 \text{adj}(2A))|$$

$$= 2^3 |2^{3-1} \text{adj}(\text{adj}(2A))|$$

$$= 2^3 \cdot (4)^3 |\text{adj}(\text{adj}(2A))|$$

$$= 2^9 |2A|^{(2)^2}$$

$$= 2^9 \cdot |2A|^4$$

$$= 2^9 \cdot 2^{12} |A|^4$$

$$= 2^{21} \cdot 2^4 = 2^{25} = (32)^n = (2)^{5n}$$

$$\therefore n = 5$$

$$\Rightarrow 3n + \alpha = 15 - 4 = 11$$

Hence this is the correct option.

13. (2)

We have been given that

$$|A| = m - n, \text{ where } 4m + n = 22 \dots\dots\dots(i)$$

$$\text{and } 17m + 4n = 93 \dots\dots\dots(ii)$$

Solving (i) and (ii)

$$m = 5, n = 2$$

$$\text{Order} = 5$$

$$\Rightarrow |A| = 3$$

We know that $|adj A| = |A|^{w-1} |adj(adj A)| = |A|^{(w-1)^2}$ where $w \times w$ is the order of the matrix.

$$\therefore \det(n \operatorname{adj}(\operatorname{adj}(mA)))$$

$$= |2 \operatorname{adj}(\operatorname{adj}(5A))|$$

$$= 2^5 |5A|^{16}$$

$$= 2^5 5^{80} |A|^{16} = 2^5 \cdot 3^{16} \cdot 5^{80}$$

$$= 3^{11} 5^{80} 6^5$$

$$\text{So, } a + b + c = 96$$

Hence this is the required option.